Highest rank for the natural alternating groups

Human Compression project: principal recovery of L044

The result recovered here is Theorem 1.1 of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1 (2016). All component proofs and their necessary finite certificates are retained. This paper checks their complete induction and source interfaces; its length alone does not measure the written replacement.

A string C-marking of rank r is an ordered family of nonidentity involutions $\rho_0, \dots, \rho_{r-1}$ generating the group, with $\rho_i\rho_j = \rho_j\rho_i$ whenever $|i-j| > 1$, and satisfying

$$\langle \rho_i : i \in I \rangle \cap \langle \rho_i : i \in J \rangle = \langle \rho_i : i \in I \cap J \rangle \quad (I, J \subseteq \{0, \dots, r-1\}).$$

The standard correspondence between such markings and abstract regular polytopes is acquired. We use the source's rank-zero convention when no generating string C-marking exists.

Theorem 1. *For $n \geq 3$, the highest rank of a string C-marking generating A_n is zero for $n = 3, 4, 6, 7, 8$, three for $n = 5$, four for $n = 9$, five for $n = 10$, six for $n = 11$, and*

$$\lfloor (n-1)/2 \rfloor \qquad (n \geq 12).$$

Proof. The complete acquired finite maxima settle every degree from three through fourteen; the values at twelve, thirteen and fourteen are $5, 6, 6$. The complete L038 existence theorem supplies a generating string C-marking of rank $\lfloor (n-1)/2 \rfloor$ for every $n \geq 12$. Its retained earlier constructions supply existence and the intersection property only: their maximality is not invoked. Thus this lower bound does not use the upper bound being recovered here.

It remains to prove the upper bound for $n \geq 15$ by strong induction. Assume it in every degree $12 \leq q < n$, retaining the exact smaller finite maxima. Let $A_n = \langle \rho_0, \dots, \rho_{r-1} \rangle$ be any string C-marking. Its noncommutation diagram is connected. Otherwise the subgroups on its components commute, intersect trivially by IP and give a nontrivial direct decomposition of the simple group A_n. Each maximal omission $G_i = \langle \rho_j : j \neq i \rangle$ is proper by independence.

If some G_i is primitive, the complete primitive reduction bounds its rank by $\lfloor (n-3)/2 \rfloor$. Hence

$$r \leq 1 + \lfloor (n-3)/2 \rfloor = \lfloor (n-1)/2 \rfloor.$$

Suppose next that a maximal omission is transitive imprimitive. Choose a maximal nontrivial block system, with m blocks of size k, so $n = km$. If $k, m > 2$, the complete large-block theorem gives the required bound. If $k = 2$, the complete pair-block theorem gives it, with its smaller-degree premises supplied by the current induction. Therefore, under a putative failure of the bound, every maximal block system of every transitive omission has exactly two blocks.

If a double omission $G_{i,j}$ inside one of these transitive omissions is transitive, the complete binary-coordinate theorem and its exact degree-sixteen certificate exclude the configuration. Its hypothesis that every transitive maximal omission preserves a binary partition has just been established; it is not assumed for an arbitrary imprimitive action. Since binary partitions have equal blocks, n is even, and the large-degree application has $n \geq 16$, as required. If instead all double omissions inside a transitive G_i are intransitive, the complete binary-omission assembly gives the desired bound. Its induction hypothesis concerns only degrees $12 \leq q < n$. Thus every transitive maximal omission is excluded under a putative rank failure.

It remains that every G_i is intransitive. An actual edge of ρ_i is eligible when it joins different G_i-orbits. Every label has such an edge: otherwise adjoining ρ_i would not make its omission transitive. If some label has exactly one eligible edge, the complete singleton and unique-crossing assembly gives $r \leq \lfloor (n-1)/2 \rfloor$. Its strict smaller-degree hypotheses are supplied by the induction assumption, not by Theorem 1 at degree n.

Otherwise each label has at least two eligible edges. Choose two per label. The complete paired count gives $2r \leq n$. In the equality case, the acquired normal form and connectedness theorem, followed by the complete paired-ladder proof, produce a partition into pairs preserved by every actual generator. Since $n \geq 15$ and equality makes n even, the ladder theorem's condition $n = 2r > 8$ holds. This contradicts the natural primitivity of A_n. Therefore $2r < n$, which is exactly the required bound. Every possibility for the maximal omissions has been exhausted, and the induction is complete. □

Attribution and retained inputs. Theorem 1 is the historical result of the four source authors. Their paper attributes the small maxima to Fernandes–Leemans–Mixer (2012), and records the large-degree bound as the solution of their earlier conjecture. No new solution of that already settled conjecture is claimed here. The lower existence constructions remain attributed through the complete L038 recovery. Classical inputs such as Whiston's generator bound, Maróti's primitive-group theorem and Conder's order bound are retained where their actual component proofs use them. Required finite computations and the completeness of the relevant GAP libraries remain declared premises, with exact execution records.

The present recovery includes new intermediate orbit, support, parity, pairing and centralizer arguments, identified separately in the component manuscripts; no priority is claimed for their elementary ingredients. The literal arbitrary-independent claim in source Proposition 3.3 is false. It is replaced only by the complete corrected theorem for independent involutory strings, with its explicit counterexample and bounded certificates retained. The finer individual ranks in the source's Table 1 are not premises or additional recovered claims of this principal-theorem contract. The mathematical assembly is distinct from publication approval, formal verification and a numerical compression audit.